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No evidence of colour-based microhabitat choice in a polymorphic marine snail

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ABSTRACT

Microhabitat choice has been proposed as a mechanism contributing to the maintenance of colour polymorphism in marine gastropods, based on observed associations between shell colour and microhabitat in certain species (e.g. *Littorina saxatilis*). To examine this hypothesis, different colour morphs of *L. saxatilis* were studied following a mark-recapture experimental design to assess whether this species preferentially chooses those microhabitats matching its shell colour, presumably favouring crypsis. Our results revealed no significant association between shell colour and microhabitat. Furthermore, despite movement patterns being affected by habitat heterogeneity, no clear evidence supporting an active search for crypsis was found. These results suggest that microhabitat choice does not play an important role in maintaining intrapopulation shell colour polymorphism in *L. saxatilis*, highlighting the relevance of alternative, more plausible processes such as differential predation.

INTRODUCTION

Natural selection is considered the primary mechanism explaining the adaptation of organisms to their environment (Futuyma, 2009). The efficacy of this process stems from the fact that certain phenotypic variants, which are genetically determined, improve the fitness of an individual with respect to alternative variants. However, natural selection may not be the sole mechanism contributing to the adaptation process. For instance, theoretical studies have shown that habitat choice and habitat heterogeneity may enhance the probability of maintaining a stable polymorphism (García-Dorado, 1986; reviewed by Hedrick, 1986; Jaenicke & Holt, 1991). Recent research also suggests that organisms may actively contribute to their adaptation by modifying their environment through niche construction or habitat choice (Odling-Smee, Laland & Feldman, 2003; Laland & O'Brien, 2010; Webster *et al.*, 2012; Camacho *et al.*, 2020). A particular mechanism that has gained attention in recent years is the possibility that individuals can directly evaluate and choose the most suitable habitat for their phenotype, referred to as 'matching habitat choice' (MHC, *sensu* Edelaar, Siepielski & Clobert, 2008). This mechanism does not require a genetic basis and can be defined as an indirect effect of an ecological trait that enables an individual to select an environment that optimizes its fitness (Edelaar *et al.*, 2008).

Recent studies demonstrate how this mechanism contributes to adaptive processes in different organisms. For instance, the grasshopper *Sphingonotus azurescens* has been reported to adjust its habitat preferences to enhance camouflage depending on its body colouration (Camacho & Hendry, 2020). This case illustrates that this mechanism may contribute to adaptive-like processes across different species when properly tested. In an example like this, the differential choice of each specimen based on its colour may have a genetic basis; however, the same genetic trait can influence both choices, and thus the proponents of this mechanism decouple genetic variability from the mechanism itself (Edelaar, Siepielski & Clobert, 2008).

The most suitable conditions for testing habitat choice or MHC may arise when a polymorphism coexists within a population, or when such polymorphism follows an environmental gradient (i.e. a cline, *sensu* Barton & Hewitt, 1985). In these scenarios, polymorphisms may be maintained by a variety of evolutionary mechanisms, predominantly natural selection, but potentially also by less frequently studied mechanisms such as habitat choice or MHC (Gefaell, Galindo & Rolán-Alvarez, 2023). Colour polymorphisms are common in marine organisms, particularly in gastropods (Gefaell *et al.*, 2023). A striking example of different colour polymorphisms is exhibited by *Littorina saxatilis*, an intertidal gastropod that inhabits North Atlantic rocky shores (Rolán-Alvarez, Austin & Boulding, 2015) and has been widely used a model organism in evolutionary studies (Johannesson, 2016; Gefaell *et al.*, 2023). This species is gonochoric, exhibits internal fertilization and undergoes direct development (i.e. ovoviviparous; Reid, 1996). These life-history characteristics are partially responsible for the striking

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capability of *L. saxatilis* to become locally adapted to different habitats and conditions.

A spectacular clinal colour distribution has been observed in *L. saxatilis* along the rocky shores of various Galician rias (inundated river valleys) in the northwest Iberian Peninsula, with different colour morphs towards both extremes of the cline (Fisher-Piette & Gaillard, 1961; Fisher-Piette, Gaillard & Jouin, 1961; Sacchi, 1979; Gefaell *et al.*, 2024b). Furthermore, the overall association of a fawn-like shell colour morph *(fulva)* with wave-sheltered habitats (generally at the interior region, near the river mouth) and a darker lineated morph (*lineata*) with wave-exposed locations has been reported to remain relatively stable for the last decades in the Ria de Vigo (the southernmost of the Rias Baixas) (Gefaell *et al.*, 2024b; Fig. 1A). Other less frequent colour morphs are found mainly at the intermediate (moderately sheltered) region, although some of the paler morphs tend to also co-occur with *fulva* towards the interior of the Ria (Gefaell *et al.*, 2021, 2024b), where intrapopulation colour polymorphism has been suggested to be maintained through negative frequency-dependent sexual selection (Gefaell *et al.*, 2024a).

These morphs conform a high diversity of shell colours, including white (*albida*), yellow (*lutea*), orange (*aurantia*), brown (*fusca*) or black (*nigra*) (colour nomenclature *sensu* Sacchi, 1979) (Gefaell *et al.*, 2024b). Interestingly, some of these morphs are suggested to be more frequent in rock substrates with similar colouration (e.g. *lineata*; Gefaell, Vigo & Rolán-Alvarez, 2025; and *lutea*, *aurantia* and *nigra*; E. Rolán-Alvarez, personal observation). Previous studies in other regions have reported shell and substrate colour to covary: in Wales, red-shelled *L. saxatilis* specimens were typically found on red rocks of the northwest and southeast coasts (Heller, 1975), while grey-shelled individuals predominantly appeared on grey rocks of the southwest coast (Byers, 1990). Irrespective of shell colouration, microhabitat choice is known to contribute to the maintenance of a well-known polymorphism in *L. saxatilis*, the 'crab' and 'wave' ecotypes. These ecotypes have adapted to different shore levels and microhabitats via divergent selection (Rolán-Alvarez, 2007; Butlin *et al.*, 2014). The 'crab' ecotype is generally restricted to the barnacle belt along the upper shore, while the 'wave' ecotype lives preferentially on the mussel belt at the lower shore (Otero-Schmitt *et al.*, 1997; Erlandsson, Rolán-Alvarez & Johannesson, 1998). These preferences have involved different adaptations to predation by crabs and wave exposure: the larger, thicker shell with a relatively small aperture of the 'crab' ecotype provides protection against crab attacks on the upper shore, while on the lower shore, in the absence of crabs, the 'wave' ecotype, with its smaller, thinner shell, has evolved a larger aperture, which has enables the foot area to be maximized, for a stronger grip against wave action (Rolán-Alvarez, 2007; Butlin *et al.*, 2014). Microhabitat choice has also been considered as a thermoregulatory strategy in other littorinids (Ng *et al.*, 2017; Moisez, Spilmont & Seuront, 2020). Therefore, habitat choice and, potentially, MHC could serve as promising mechanisms for mediating adaptation to different habitats in *L. saxatilis*.

In this study, we aim to investigate whether habitat choice, a necessary pre-requisite for MHC, could potentially contribute to the maintenance of intrapopulation shell colour polymorphisms in marine gastropods, through a combination of habitat choice and background natural selection. We selected two polymorphic populations of *L. saxatilis* from intermediate locations from the Ria de Vigo and carried out a mark-recapture experimental design to test the preference of the colour morphs for a specific microhabitat.

MATERIAL AND METHODS

Animal collection, maintenance and experimental design

Littorina saxatilis individuals were collected from two polymorphic populations within the intermediate, moderately-sheltered zone of the colour cline in the Ria de Vigo (Gefaell *et al.*, 2024b): Canabal (north coast; 42°15′45.9″N, 8°45′26.3″W) and Portiño (south coast; 42°10′06.4″N 8°49′02.8″W) (Fig. 1A, B, G). These populations display similar frequencies of the *lineata* morph to other colour morphs, presumably inhabiting colour-matching substrates. The characteristic orange granitic rock substrate of Canabal (Fig. 1C) corresponds with a particularly high frequency of the *aurantia* (orange) morph (Fig. 1E). Furthermore, *lineata* individuals (Fig. 1F) are often found on dark patches of the black lichen *Lichina pygmaea* (Fig. 1D). Regarding the other locality, Portiño, yellow granitic rock substrate (Fig. 1H) and barnacle-covered rocks (Fig. 1I) might be mostly occupied by the *lutea* (yellow) and *lineata* morphs, respectively (Fig. 1J, K). A mark-recapture experiment was conducted for each locality, following a reciprocal transplant design that has been previously employed in *L. saxatilis* (see Rolán-Alvarez, Johannesson & Erlandson, 1997).

The first experiment was performed in Canabal during 3 consecutive days in March 2023. A total of 20 sites (10 with bare orange rock and 10 with black lichen patches) were selected and marked using a 23 mm diameter circular yellow plastic circle screwed into the rock during low tide. The area (cm^2) of each site was estimated by measuring the distance (cm) from the mark (release point hereafter) to the beginning of the other habitat in four directions. Usually, these pure sites (homogeneous patches) were surrounded by both microhabitats. Then, a sample of 60 *aurantia* and 60 *lineata* snails was collected and transported to the laboratory in plastic boxes with wet cardboard and algae to mimic their natural environment. Once in the laboratory, the shell size of all individuals (i.e. columella length) was measured using vernier calipers and the presence or absence of shell scars was noted (old scars on the shell indicate crab predation attempts, following Butlin *et al.*, 2014; Gefaell *et al.*, 2025). We selected sizes over 7 mm to minimize immature snails, which could present different behaviour compared to adults (Johannesson, Rolán-Alvarez & Ekendahl, 1995). A tag number, identifying both individual and release point, was glued (UHU Power Glue Gel) onto each shell. All specimens were maintained at 5 °C for 16 h in plastic boxes with wet cardboard until their release on day 2. Each release point included 3 individuals of each colour morph (*aurantia* and *lineata*), alternating positions to minimize spatial effects. The snails were placed in the release points and recaptured the third day after one tidal cycle (*c.* 12 h). For each recaptured snail we recorded the tag number, the distance (cm) between the snail and the release point, the recapture habitat (bare rock or black lichen patch), whether the snail changed its habitat (i.e. the recapture habitat differs from the release habitat) and the direction of movement (towards land or the sea). All the recaptured snails were brought to the laboratory and euthanized at −20 °C prior to dissection and sexing.

The second experiment was carried out in Portiño in May 2024, following a similar mark-recapture procedure to the previous experiment. To compare *lutea* and *lineata* colour morphs, a circular area (25 cm radius) around 10 release points was manipulated to obtain both habitats (bare yellow rock and barnacle-covered). This area was divided into four equal parts and noncontiguous parts were then cleared by removing all the barnacles with a chisel, corresponding to the bare rock habitat. This design was chosen in this sampling site because the two microhabitats could be easily distributed by removing the barnacles from certain parts of the rock, a strategy that could not be followed in Canabal with the black lichen microhabitat. This experiment was divided into two sessions of 3 consecutive days. For each session, a sample of 40 individuals of each colour morph (*lutea* and *lineata*) were collected and transported to the laboratory for phenotyping and individual tagging, as described above. The snails were released on day 2 in groups of 4 individuals of each colour morph per release point, alternating positions to minimize spatial effects. On day 3 (*c.* 24 h later; two tide cycles, as the first low tide was too early), snails were recaptured and, as described above, for each we recorded the tag number, the distance (cm) between the snail and the release point, the recapture habitat (bare yellow rock or barnacle-covered) and the di-

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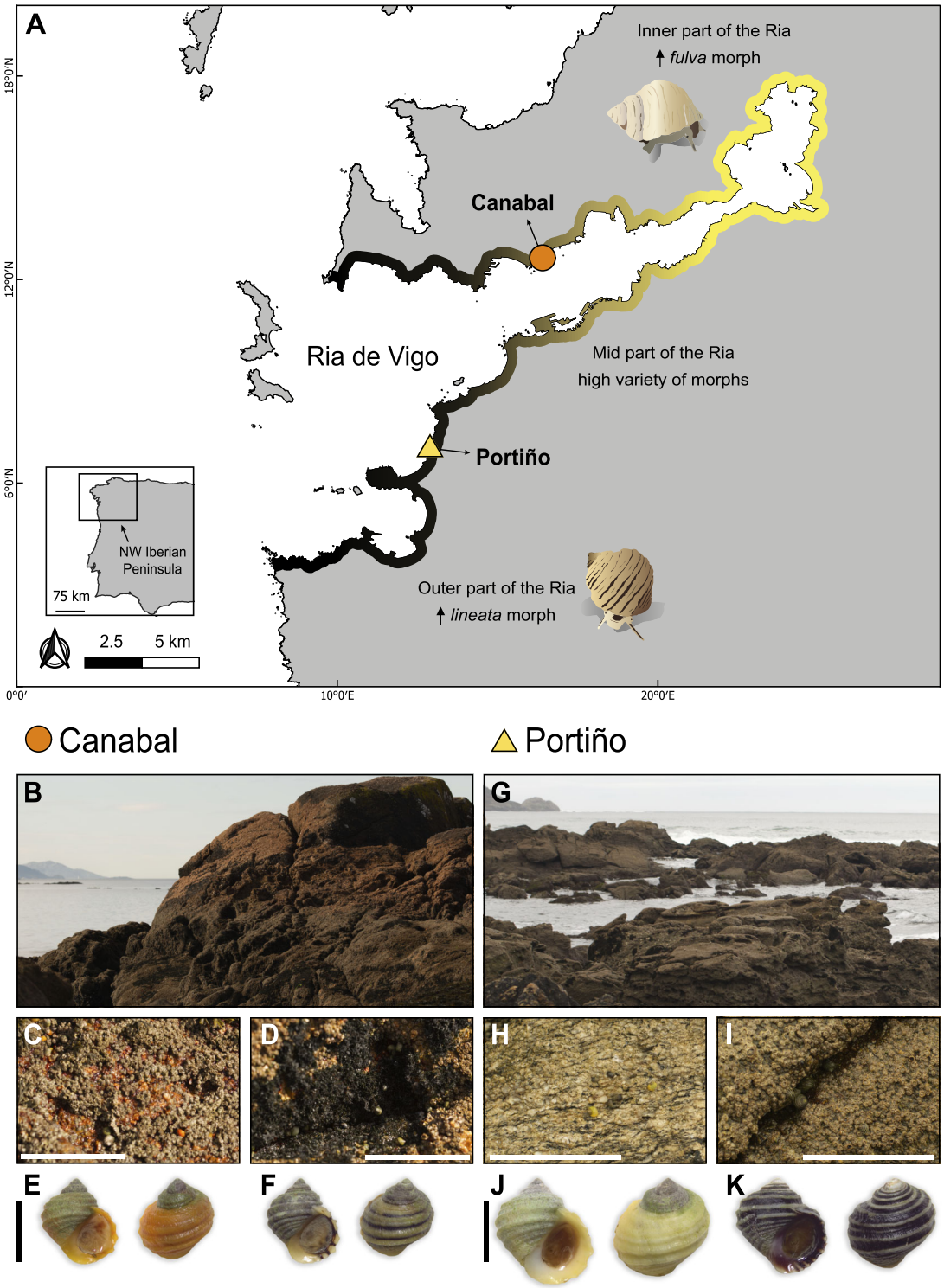


Figure 1. Study area and *Littorina saxatilis* colour morphs. **A.** Geographical reference of the Ria de Vigo and study sites (Canabal and Portiño) within *L. saxatilis* colour cline, illustrated as a gradual colour variation from black (*lineata* morph, dominating the outer region) to light fawn colour (*fulva* morph in the interior region) along the coastline of the Ria (based on Gefaell *et al.*, 2024b). Maps were created using shapefiles from Instituto Geográfico Nacional for the Rias Baixas (BDLJE CC-BY 4.0, www.ign.es) and Natural Earth for the Iberian Peninsula (www.naturalearthdata.com) datasets in QGIS (qgis.org), then modified in Affinity Designer 2 (www.affinity.studio). **B–F.** Microhabitats and colour morphs studied in Canabal (experiment 1). **B.** Photograph of the location. **C.** Bare rock substrate and with *aurantia* colour morph. **D.** Black lichen patch with *lineata* colour morph. **E.** Shell of *aurantia* colour morph. **F.** Shell of *lineata* colour morph. **G–K.** Microhabitats and colour morphs studied at Portiño (experiment 2). **G.** Photograph of the location. **H.** Bare rock substrate with *lutea* colour morph. **I.** Barnacle-covered rock with *lineata* colour morph. **E.** Shell of *lutea* colour morph. **F.** Shell of *lineata* colour morph. White scale bars = 50.0 mm; black scale bars = 5.0 mm. All the images were taken in RAW format (original by R.V., using a Nikon D500 digital camera and a Sigma 105 mm f/2.8 macro lens) and calibrated using a white balance reference card to standardize and compare both shell and substrate colours (Stevens *et al.*, 2007).

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Table 1. Results of contingency G tests for experiment 1, presenting *G* value, degrees of freedom (df) and probability (*P* value), as well as the output of each power analysis with its effect size (*ES*), sample size (*N*), the statistical power of the test with the real sample size (*Power*) and the minimum sample size required to achieve a reliable model (i.e. *Power* > 0.8; N_e for 0.8).

Factor	Predictor	*G*	df	*P* value	*ES*	*N*	*Power*	N_e for 0.8
Recapture	Scar presence	0.2	1	0.663	0.039	120	0.07	5,171
	Colour morph	0.06	1	0.810	0.022	120	0.06	16,318
Scar presence		24.7	1	<0.001***	0.417	120	0.99	46
Sex		1.7	2	0.431	0.129	99	0.19	582
Recapture habitat		0.3	1	0.585	0.055	99	0.08	2,611
Habitat change		0.003	1	0.960	0.005	99	0.05	30,3830
	Release habitat	12.3	1	<0.001***	0.337	99	0.92	70

*$P < 0.05$, **$P < 0.01$, ***$P < 0.001$.

rection of the movement (towards land, the sea or an intermediate route). Finally, any tag was successfully removed since no effect of sex was detected in the first experiment (i.e. no dissection needed), and all recaptured snails were returned alive to their respective habitat.

Statistical analyses

The association between qualitative variables was evaluated by a contingency G test. For Canabal (experiment 1), these variables included 'colour morph' (fixed, two levels: *aurantia*, *lineata*), 'scar presence' (0 = absent, 1 = present), 'sex' (fixed, three levels: male, female, immature), 'release habitat' (fixed, two levels: bare rock, black lichen patch), 'recapture' (fixed, two levels: 0 = lost, 1 = recaptured), 'recapture habitat' (fixed, two levels: bare rock, black lichen patch), 'habitat change' (fixed, two levels: same, different) and 'route' (fixed, two levels: land, sea, intermediate). These comparisons were also performed for Portiño (experiment 2) using the same variables, except for 'release habitat' and 'habitat change' given its different experimental design. Furthermore, 'colour morph' included *lutea* level instead of *aurantia*. The statistical reliability of all these comparisons was examined using power analyses for each case following Cohen (1988). Exploratory analyses to explain the factors contributing to the distance moved by recaptured snails from their release point were conducted by a stepwise multiple regression, using as predictors all qualitative variables mentioned, as well as 'release habitat mean distance', 'release habitat distance variance', 'release habitat area' for experiment 1 and 'shell size' for both experiments. Statistical power was evaluated for relevant predictors as well. Differences in shell size between colour morphs were assessed by a nonparametric Kruskal–Wallis test (Sokal & Rohlf, 1995), as well as the area of the release habitats from Canabal (Experiment 1). All statistical analyses were performed using 'stats', 'DescTools' and 'pwr' packages in R v. 4.4.2 (R Core Team, 2024).

Ethical standards

All procedures involving the collection and use of *L. saxatilis* specimens were conducted in accordance with the principles outlined in European Union Directive 2010/63/EU on the protection of animals used for scientific purposes and complied with Spanish Law 65/2013. Specimen sampling and euthanasia were carried out with minimal impact on natural populations and following established ethical guidelines to ensure animal welfare.

RESULTS

Experiment 1: orange bare rock against black lichen patches

From a sample of 120 (60 of each colour morph) marked and released snails, 82.5% were successfully recaptured. Shell colouration of the snail was not associated with being recaptured or not by a G association test (Table 1), and there were no shell size differences between colour morphs (Kruskal–Wallis test: $\chi^2 = 1.6$, df = 1, $P = 0.207$). Although shell colouration was strongly associated with scar presence (Table 1; statistical power > 0.8) and the percentage of scars within the sample was relatively high (17.5%), neither scars nor colour morph determined whether a snail was recaptured or not (Table 1). Indeed, we found no effect of colour morph on whether a snail was recaptured on bare rock or a black lichen patch, or it changed from one habitat to another (Table 1). The snails moved a mean distance of 19.8 ± 22.9 (SD) cm per snail (*c.* 1.7 cm per hour) from their release point to the recapture point, and this distance depended on whether a snail changed its habitat or not using a stepwise multiple regression model (Table 2; statistical power = 1 based on 1,000 simulations with the real $N = 99$; a sample of $N = 24$ snails would be enough to consistently detect this effect). The mean distance was 13.0 ± 14.5 cm for those remaining in the same habitat in which they were released ($n = 79$ individuals) and 46.4 ± 30.2 cm for those who were recaptured within a different habitat ($n = 20$ individuals) (see distances by colour morph in Fig. 2). Distance differences between release habitats were also found by stepwise multiple regression (Table 2; statistical power = 0.82 based on 1,000 simulations with the real $N = 99$), and whether a snail was released in one or another habitat strongly affected the preference of changing its habitat (Table 1; statistical power > 0.8). However, the quadrats of black lichen patches were significantly smaller (1,773 ± 872 cm^2) than those with bare rock (3,348 ± 1,338 cm^2) (Kruskal–Wallis test: $\chi^2 = 36.6$, df = 1, $P < 0.001$).

Experiment 2: yellow bare rock against barnacle-covered rock

For experiment 2, only 46.2% of total snails were recaptured (session 1 = 52.5%, session 2 = 40.0%). Again, there was no significant association between colour morph and being recaptured or not in any session or both pooled by a G association test (Table 3). For recaptured snails, only 5.4% displayed scars, and scar frequency did not differ between colour morphs (Table 3), as well as shell size (Kruskal–Wallis test, $P > 0.05$ in each session and both pooled). Consistent with the results for experiment 1, whether recapturing a snail from bare rock, barnacle-covered habitat or another habitat (9.5% of the snails in session 1 were recaptured on black lichen patches) did not depend on shell colouration (Table 3). All these comparisons, however, were insufficiently reliable to detect a minimal effect in case it existed (statistical power < 0.2 in all cases, with a low effect size as well; Table 3). The average distance per snail from the release point (112.0 ± 94.1 cm) was larger than in experiment 1 (19.8 ± 22.9 cm), which represents a net movement of *c.* 4.7 cm per hour, although it could not be explained by any of the independent variables for the whole experiment by a stepwise multiple regression (Supplementary Material Table S1). When analysing ses-

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Table 2. Result of the stepwise multiple regression to explain variability in the distance moved by the snails from the release point to the recapture point for experiment 1.

Predictor	Level	Estimate (β)	SE	*t* value	*P* value
Intercept		17.573	2.619	6.779	$<0.001^{***}$
Habitat change (reference: Same habitat)	Different habitat	37.959	4.792	7.921	$<0.001^{***}$
Release habitat (reference: Bare rock)	Black lichen patch	− 10.955	3.850	− 2.845	0.0054^{*}

$R^2 = 0.39$; $F_{2,96} = 31.39^{***}$; $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$.
The model included 'colour morph', 'shell size', 'scar presence', 'sex', 'release habitat', 'release habitat mean distance', 'release habitat distance variance', 'release habitat area', 'recapture habitat', 'habitat change' and 'route' variables as the predictors. Coefficient estimate (β), standard error (SE), *t* value and its probability (*P* value) shown for each level of the output predictors. Also provided the multiple R^2 coefficient and the F-statistic of the model.

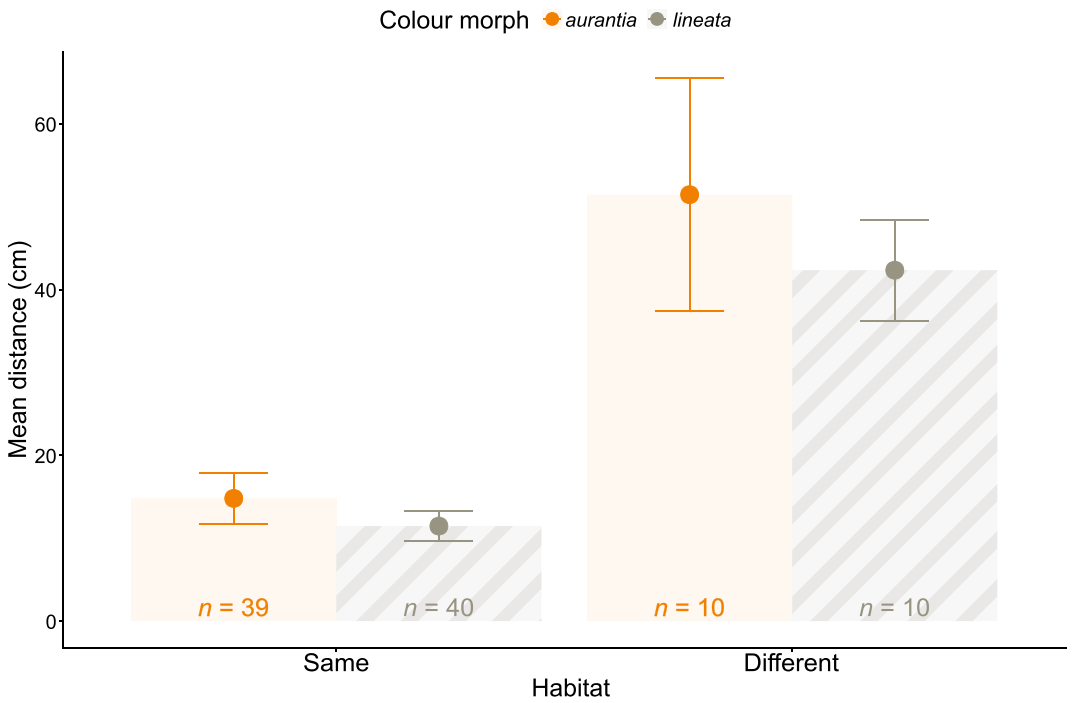


Figure 2. Mean distance (cm) from the initial point to the point where snails were recaptured (i.e. in the same habitat they were released in or a different habitat) by colour morph in Canabal (experiment 1). Error bars indicate standard error.

Table 3. Results of contingency G test comparing the qualitative variables 'recapture', 'scar presence' and 'recapture habitat' with 'colour morph' (*lutea* or *lineata*) as the predictor.

Factor	Predictor	Session	*G*	df	*P* value	*ES*	*N*	*Power*	N_e for 0.8
Recapture	Colour morph	1	0.80	1	0.370	0.100	80	0.15	783
		2	0.21	1	0.648	0.051	80	0.07	3,014
		Pooled	0.10	1	0.751	0.025	160	0.06	12,488
Scar presence		1	0.19	1	0.664	0.066	42	0.07	1,784
		2	1.55	1	0.213	0.191	32	0.19	215
		Pooled	0.99	1	0.319	0.113	74	0.16	614
Recapture habitat		1	0.05	2	0.975	0.035	42	0.05	7,883
		2	0.47	1	0.491	0.122	32	0.11	532
		Pooled	0.23	2	0.896	0.054	74	0.07	74

$^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$.
G value, degrees of freedom (df) and probability (*P* value) are provided for each session and both pooled, as well as the output of each power analysis with its effect size (*ES*), sample size (*N*), the statistical power of the test with the real sample size (*Power*) and the minimum sample size required to achieve a reliable model (i.e. *Power* > 0.8; N_e for 0.8).

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Table 4. Result of the stepwise multiple regression to explain the variability in distance moved by the snails from the release point to the recapture point for session 1 of experiment 2.

Predictor	Level	Estimate (β)	SE	*t* value	*P* value
Intercept		71.23	18.49	3.852	$<0.001^{***}$
Recapture habitat (reference: Bare rock)	Barnacle-covered	60.12	29.31	2.051	0.047^{*}
	Black lichen patch	124.47	49.09	2.535	0.016^{*}
Scar presence (reference: Absent)	Present	− 98.80	54.85	− 1.801	0.080

$R^2 = 0.19$; $F_{4,38} = 3.036^{*}$, $P = 0.041$; $^{*}P < 0.05$, $^{**}P < 0.01$, $^{***}P < 0.001$.
The model included 'session', 'recapture habitat', 'colour morph', 'shell size', 'scar presence' and 'route' as the predictors. Coefficient estimate (β), standard error (SE), *t* value and its probability (*P* value) shown for each level of the output predictors. Also provided are the multiple R^2 coefficient and the F-statistic of the model.

sions separately, distance was associated only with recapture habitat in session 1 when including those snails recaptured in black lichen patches (Table 4; statistical power = 0.7 based on 1,000 simulations with the real $N = 42$). Those snails that were recaptured in bare rock substrate moved a shorter distance on average (71.2 ± 84.7 cm, $n = 22$) than the snails recaptured on barnacle-covered rocks (119.0 ± 98.5 cm, $n = 16$) or black lichen patches (171.0 ± 68.1 cm, $n = 4$).

DISCUSSION

In this work, we assess microhabitat choice in *Littorina saxatilis* populations displaying intrapopulation shell colour polymorphism, a necessary condition prior to formally studying MHC in this species. MHC represents an understudied mechanism that could potentially contribute to explain these patterns of shell colour variation within the intermediate zone of the colour cline (Ria de Vigo). However, it is essential that we evaluate any potential conscious decision from the organism (Edelaar *et al.*, 2008). Previous studies have shown that marine snails exhibit the capacity to actively choose between different microhabitats in the pursuit of a more suitable temperature (Ng *et al.*, 2017; Moisez, Spilmont & Seuront, 2020) or degree of wave exposure (Otero-Schmitt *et al.*, 1997; Erlandsson *et al.*, 1998; Rolán-Alvarez, 2007). In contrast, our results from the microhabitat choice experiments suggest that shell colour morphs of *L. saxatilis* do not discriminate between the natural microhabitats assessed here (i.e. *aurantia vs lineata* colour morphs between orange bare rock and black lichen patches, and *lutea vs lineata* between yellow bare rock and barnacle-covered rock). We will first discuss certain potential limitations of our experimental design and, then, explore whether the results we have obtained are likely to be representative of the biology of *L. saxatilis*. Finally, we will highlight the potential relevance of additional findings, such as slight differences in the distances covered by the snails according to their recapture microhabitat.

The potential for predation to mask the pattern of movement among different morphs, particularly when the differences are colour-based, as suggested by Gefaell *et al.* (2025) for this species, is a salient concern to consider in these experiments. Accordingly, installing predation cages in the field is an extensively utilized method for restricting access by predators (Ekendahl, 1998; Johannesson & Ekendahl, 2002; Hooks & Padilla, 2014). In our study, the complex topography of the experimental area made it difficult to find places flat enough to include various microhabitats within a single cage. Furthermore, our study was shown to lack power for most of the variables we tested, requiring unreasonable sample sizes for these populations regardless of the inclusion of predator cages. An alternative approach would be a mesocosm experiment that excludes predation, despite its reduced ecological representation of a natural scenario. However, the high recapture rates (especially in Canabal) and the absence of evident predation marks, asobserved by us, suggest a reduced influence of predation in our study. Gefaell *et al.* (2025) conducted a mark-recapture experiment to assess the role of crypsis in monomorphic *lineata* populations at the exposed region of the Ria de Vigo. Although the tethering method followed in that experiment excluded the chance of missing snails due to movement, they observed low predation rates after >1 week (*c.* 1% per day in average), and this is consistent with the minimal predation influence we found after only 1–2 tide cycles. Nonetheless, if such an effect exists, it would require >5,000 snails to be detected in the field according to our power analysis. Therefore, it is likely that the unrecaptured snails either hid in crevices or left the experimental area. Another potential limitation of our experimental design is that we have not discriminated between microhabitat colour and structure. For instance, the dark natural substrate is not just different because it is dark, but also because it is lichen-covered. However, such a consideration may be more relevant if the results had been positive. In the case of our study, snails were unable to distinguish between microhabitats based on their own shell colour, regardless of whether we focused on colour or structure.

In Canabal, given that *lineata* morph is darker than *aurantia*, it should be less visible on the black lichen patches, while the opposite may occur in bare rock microhabitats. The existence of such differential crypsis may suggest *a priori* a certain net advantage for *aurantia* due to the reduced presence of black lichen patches at this location (E. Rolán-Alvarez, personal observation). Indeed, the observed prevalence of shell scars was slightly higher in *lineata* specimens (33%), suggesting that predation by crabs would affect this morph in particular (Butlin *et al.*, 2014). However, despite being less cryptic than *aurantia*, the ability to selectively occupy distinct microhabitats may not have evolved in *lineata*. This could be due to the absence of colour vision in this snail species, in line with most gastropods (Gefaell *et al.*, 2023). Consequently, it may be challenging for them to identify correlated traits that would allow them to find a presumably more adequate microhabitat. Furthermore, it is plausible that such differences may also vary geographically or temporally, prompting uncertain selection pressures that do not facilitate the fixation of a given allele in the population.

Similarly, in Portiño, the *lineata* morph may be more cryptic on barnacle-covered rocks (Johannesson *et al.*, 1993; Rolán-Alvarez, 2007; Butlin *et al.*, 2014), while yellow specimens should be more difficult to locate on yellow bare rock. In this case, the absolute frequency of shell scars was low, and no differences between colour morphs were detected. Such results are probably attributable to a reduced frequency of crab predation in this location and the coexistence of both microhabitats at rather similar frequencies (E. Rolán-Alvarez, personal observation). Moreover, neither of the two colour morphs manifested a clear preference for any of the two microhabitats, thereby reiterating the absence of empirical evidence in support of the microhabitat choice or MHC hypotheses. Although MHC could be a promising new mechanism that may in some way facilitate adaptation (Edelaar *et al.*, 2008; Dreiss *et al.*, 2012), the present study suggests that this may not be the case in *L. saxatilis*, at least with respect to shell colouration.

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A salient finding that merits further discussion is the differential mobility (per hour) observed between the two experimental locations: snails moved nearly 4 times the distance in Portiño than in Canabal. This outcome can be partially explained by recapture occurring after two tide cycles in Portiño, but only one in Canabal. It is plausible that part of the first tide cycle may be necessary to recover from any stress induced by manipulation in the laboratory, thus potentially affecting snail activity after release. Although distance was not related to shell colouration, we found that snails recaptured in a new microhabitat travelled further. However, this observation provides no additional insight into movement patterns based on microhabitat choice since the area of the release microhabitats was also significantly different, at least in Canabal. For Portiño, this tendency was observed only in the second session, in which 4 individuals (9.5%) were recaptured much further from their release point than the rest of the snails on a black lichen patch. In summary, our results show that the snails moved larger or shorter distances independent of shell colour or the two microhabitats studied at each location. It must be noted, however, that our findings do not imply an absence of microhabitat preferences in this species during movement, feeding or searching for refuge (see, e.g. Otero-Schmitt *et al.*, 1997; Erlandsson *et al.*, 1998), but rather that they do not reveal a different preference based on colour morphs. Indeed, the 'crab' and 'wave' ecotypes known to exist in *L. saxatilis* exhibit strong microhabitat preferences, inhabiting the upper and lower shore, respectively, which directly corresponds with the conditions under which they show higher survival (Rolán-Alvarez, 2007; Rolán-Alvarez *et al.*, 2015). Of course, natural selection may still contribute to the maintenance of these shell colour polymorphisms via differential (predatory) selection on different microhabitats, but such a process may be disrupted by coloured specimens moving unpredictably across microhabitats on a day-to-day basis. Despite the inherent limitations of our study, we contend that our findings point to a minimal role for microhabitat choice or MHC in the maintenance of colour diversity in this or similar species.

SUPPLEMENTARY MATERIAL

Supplementary material is available at *Journal of Molluscan Studies* online.

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CONFLICT OF INTEREST

The authors declare that they have no conflict of interest.

DATA AVAILABILITY

All data and analysis codes are available at https://github.com/ramonvigoq/MicrohabitatChoice_ColorPolymorphism.

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